ECON 381: Homework 3 Solutions

Fall 2026 — Instructor solution key

Source. Exercises adapted from James H. Stock and Mark W. Watson, *Introduction to Econometrics*, 4th edition (Pearson).

Conventions. Results use the supplied data for 17,870 workers. Empirical earnings are annual amounts in 2012 dollars. Regression standard errors are heteroskedasticity-robust. Intervals use 1.96; two-sided p-values use $2\Phi(-|t|)$. Calculations use unrounded estimates.

1 Empirical Textbook Exercises

E4.2(a): Median height

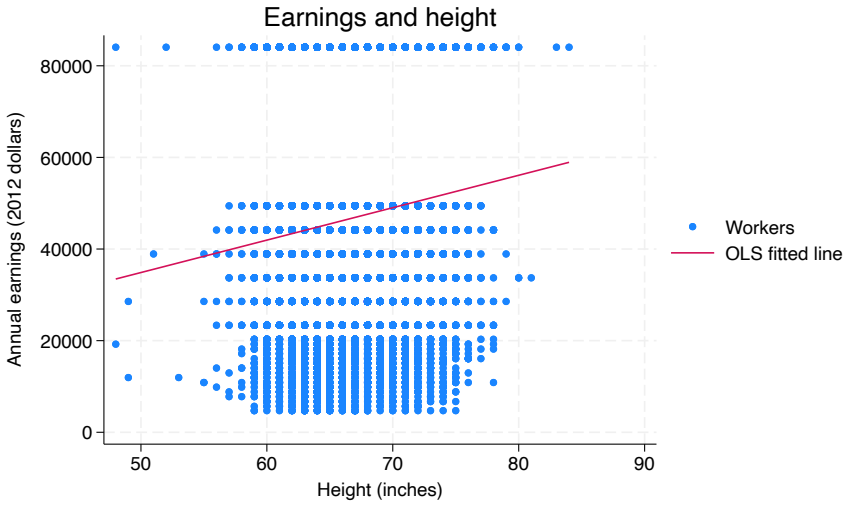
The sample median height is **67 inches**.

E4.2(b): Average earnings by height group

Height group	Number of workers	Mean annual earnings
At most 67 inches	10,114	\$44,488.44
Greater than 67 inches	7,756	\$49,987.88

These are the answers to (i) and (ii). For (iii), taller workers earn **\$5,499.44 more per year** on average, calculated by subtracting the unrounded group means. This descriptive difference does not establish a causal effect of height.

E4.2(c): Scatterplot and horizontal bands



Earnings were recorded in **23 income brackets**. Case and Paxson assigned each worker an estimated average for that worker's bracket, then converted the amounts to 2012 dollars. Identical bracket values produce horizontal bands. Recorded values are grouped representations, rather than each worker's exact earnings.

E4.2(d): Pooled regression and predictions

The estimated regression for all workers is

$$\widehat{Earnings} = -512.7336 + 707.6716\, Height, \qquad R^2 = 0.0108753.$$

(i) The estimated slope is **$707.67 per inch**: an additional inch of height is associated with about $707.67 higher predicted annual earnings.

(ii) Substitute each height into the fitted equation, using the unrounded coefficients:

Height (inches)	Predicted annual earnings
67	$46,901.26
70	$49,024.28
65	$45,485.92

E4.2(e): Measuring height in centimeters

Since $Height_{\text{cm}} = 2.54\, Height_{\text{in}}$, divide the slope by 2.54. The four answers are:

Quantity	Value using height in centimeters
(i) Slope	$707.6716/2.54 = 278.6108$ dollars per cm
(ii) Intercept	-512.7336 dollars
(iii) R^2	0.0108753 (approximately 0.0109)
(iv) SER	$26,777.24

Rescaling height preserves fitted earnings and residuals, so R^2 and SER are unchanged. Here $SER = \sqrt{\sum_i \widehat{u}_i^2/(n-2)}$ measures residual dispersion; it is distinct from a coefficient's standard error.

E4.2(f) and (g): Separate regressions for women and men

$$\text{Women:} \quad \widehat{Earnings} = 12{,}650.8577 + 511.2222\, Height,$$
$$\text{Men:} \quad \widehat{Earnings} = -43{,}130.3423 + 1{,}306.8599\, Height.$$

For **(f)(i)**, the female slope is **$511.22 per inch**. For **(f)(ii)**, a woman 1 inch taller than the female sample mean is predicted to earn **$511.22 more** than female sample-average annual earnings. For **(g)**, the corresponding male slope and predicted earnings difference are both **$1,306.86 per inch**. An OLS line with an intercept passes through the sample means within each group. Thus $\widehat{Y}(\overline{X} + 1) = \overline{Y} + \widehat{\beta}_1$, which explains each comparison with group-average earnings.

E4.2(h): Omitted factors and the conditional mean

There is no reason to conclude from this regression alone that $E[u_i \mid Height_i] = 0$. Height may be correlated with omitted determinants of earnings, including childhood conditions, family background, education, health, sex, or physical strength. Thus the slope describes an association and need not identify a causal effect of height on earnings.

E5.1(a)–(c): Statistical significance and confidence intervals

For each sample, test $H_0 : \beta_1 = 0$ against $H_1 : \beta_1 \neq 0$ using

$$t = \frac{\widehat{\beta}_1}{SE(\widehat{\beta}_1)}, \qquad CI_{0.95} = \widehat{\beta}_1 \pm 1.96\, SE(\widehat{\beta}_1).$$

The standard errors below are heteroskedasticity-robust. Slopes, standard errors, and interval endpoints are annual dollars per inch.

Sample	n	Slope	Robust SE	95% normal CI
(a) All workers	17,870	707.67	50.40	[608.90, 806.45]
(b) Women	9,974	511.22	97.58	[319.96, 702.49]
(c) Men	7,896	1,306.86	98.86	[1,113.10, 1,500.62]

Sample	Test statistic	Normal p-value	Decision at 5%
All workers	14.0425	8.56×10^{-45}	Reject H_0
Women	5.2388	1.62×10^{-7}	Reject H_0
Men	13.2197	6.75×10^{-40}	Reject H_0

The slope is positive and statistically significant at the 5% level in all three samples. Each interval excludes zero, each absolute test statistic exceeds 1.96, and each p-value is below 0.05. These tests establish statistical evidence of an association; they do not establish causality.

E5.1(e): Occupations in which strength is less important

One reasonable restriction includes occupation codes 1–5: executives/managers, professionals, technicians, sales, and administrative workers. Physical strength is less likely to be central to productivity in these occupations. Pooling these workers gives

$$\widehat{Earnings} = -2{,}149.5193 + 829.7842\, Height, \qquad n = 11{,}020.$$

The robust slope standard error is 65.1138, so

$$t = \frac{829.7842}{65.1138} = 12.7436, \qquad p \approx 3.38 \times 10^{-37},$$
$$CI_{0.95} = 829.7842 \pm 1.96(65.1138) = \boxed{[702.16, 957.41]}.$$

The positive association **does not disappear**: it remains about \$829.78 per inch and is statistically significant at 5%.

The separate regressions provide additional detail:

Occupation	n	Slope	Robust SE	Significant at 5%?
Executives/managers	1,906	469.46	153.05	Yes
Professionals	3,158	622.76	116.17	Yes
Technicians	875	649.72	213.04	Yes
Sales	1,957	1,372.38	146.44	Yes
Administration	3,124	201.22	131.46	No

Strength alone cannot explain the association. This restriction does not identify a causal effect. Other well-justified occupation restrictions are acceptable. **Part (d) is not assigned.**

2 Analytical Textbook Exercises

Exercise 5.2

(a) The estimated male-minus-female wage gap is the coefficient on *Male*:

$$\boxed{\widehat{\beta}_1 = \$2.12 \text{ per hour}}.$$

(b) Test $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$. The statistic and two-sided normal p-value are

$$t = \frac{2.12 - 0}{0.36} = 5.8889, \qquad p = 2\Phi(-5.8889) \approx 3.89 \times 10^{-9}.$$

Reject H_0 at the 5% level (and at the 1% level). The gap is statistically significantly different from zero. The p-value is extremely small, not exactly zero.

(c) The approximate 95% confidence interval is

$$2.12 \pm 1.96(0.36) = [1.4144, 2.8256] = \boxed{[\$1.41, \$2.83] \text{ per hour}}.$$

(d) For women, $Male = 0$, so the sample mean wage is **\$12.52 per hour**. For men, $Male = 1$, so the sample mean wage is

$$12.52 + 2.12 = \boxed{\$14.64 \text{ per hour}}.$$

Part (e) is omitted.

Exercise 5.8: Normal approximation with $n = 100$

(a) The intercept's approximate 95% confidence interval is

$$43.2 \pm 1.96(10.2) = 43.2 \pm 19.992 = [23.208, 63.192] = \boxed{[23.21, 63.19]}.$$

(b) Test $H_0 : \beta_1 = 55$ versus $H_1 : \beta_1 \neq 55$. Center the estimate at the null value:

$$t = \frac{\widehat{\beta}_1 - 55}{SE(\widehat{\beta}_1)} = \frac{61.5 - 55}{7.4} = 0.8784.$$

Since $|0.8784| < 1.96$, **do not reject H_0 at the 5% level**. Equivalently, the two-sided normal p-value is

$$p = 2\Phi(-0.8784) \approx 0.3797 > 0.05.$$

The reference distribution is standard normal, as required in the assignment.

(c) The sample does not provide sufficient evidence that the population slope differs from 55 at the 5% significance level. Failing to reject does not prove that $\beta_1 = 55$ and does not mean that the null has a 95% probability of being true. The 5% level refers to the procedure's approximate probability of rejecting a true null across repeated samples.

3 True/False Questions

1. **False.** Heteroskedasticity means that the conditional *variance*, $\text{Var}(u_i \mid X_i)$, varies with the regressors. It does not require different conditional means. In particular, heteroskedasticity can coexist with the least squares assumption $E[u_i \mid X_i] = 0$.

2. **False.** For a fixed dataset, OLS minimizes the same sum of squared residuals regardless of the variance assumption used for inference. Changing from homoskedasticity-only to heteroskedasticity-robust standard errors changes the estimated coefficient uncertainty, not the OLS coefficients. For example,

$$\widehat{\beta}_1 = \frac{\sum_i (X_i - \overline{X})(Y_i - \overline{Y})}{\sum_i (X_i - \overline{X})^2}$$

   uses the sample data and does not change when the standard-error formula changes.

3. **False.** Under the usual regularity conditions, a heteroskedasticity-robust standard error consistently estimates the sampling standard deviation of the OLS coefficient under homoskedasticity as well. Homoskedasticity is a special case in which the robust procedure remains valid. Robust and homoskedasticity-only standard errors need not be numerically identical in a finite sample.

4. **False.** The interval based on 1.96 relies on a large-sample normal approximation. With $n = 10$, the stated least squares assumptions do not guarantee that this approximation is accurate or that the interval attains 95% coverage. Coverage is a repeated-sampling property; a particular realized interval either contains the fixed parameter or does not.

Companion files and reproduction

- `ECON381_HW3_Solutions.do`: the complete Stata code, with answers and interpretations in comments.
- `ECON381_HW3_Solutions.log`: the completed Stata run, including the normal-approximation calculations.

To rerun the analysis, set Stata's working directory to the Assignment 3 folder and run:

```
do "ECON381_HW3_Solutions.do"
```

The data description credits Anne Case and Christina Paxson, "Stature and Status: Height, Ability, and Labor Market Outcomes," *Journal of Political Economy*, 2008, 116(3): 499–532.